MAJOR MOON

Titan

DISTANCE FROM SATURN 758,073 miles (1.22 million km)

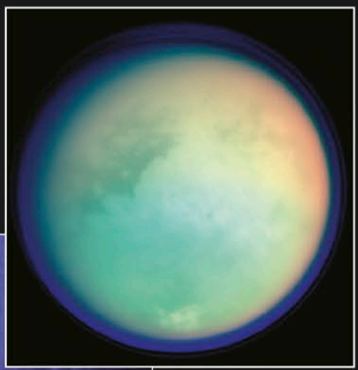
ORBITAL PERIOD 15.95 Earth days

DIAMETER 3,200 miles (5,150 km)

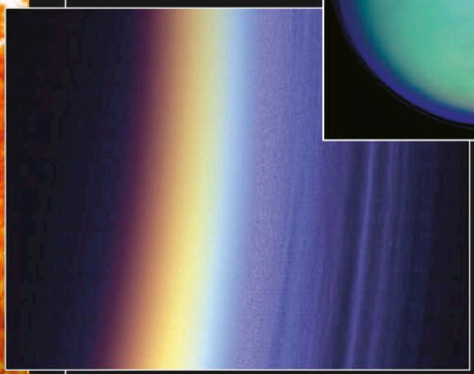
Titan was discovered in 1655 by the Dutch scientist Christiaan Huygens. It is the second-largest moon in the solar system after Jupiter's Ganymede (see p.186) and is by far the largest of Saturn's moons. This Mercury-sized body is also one of the most fascinating. A veil of smoggy haze shrouds the moon and permanently obscures the world below. Titan is intriguing, not least because the chemistry of the atmosphere has similarities to that of the young Earth before life began. The first chance to see the surface and test the atmosphere came in 2005, when Cassini turned its attention to Titan, and Huygens plunged through the atmosphere to the surface (see panel, right).

The nitrogen-rich atmosphere extends for hundreds of miles above Titan. Layers of yellow-orange, smoglike haze high within it are the result of chemical reactions triggered by ultraviolet light. Methane clouds form much closer to the surface. These rain methane onto Titan, where it forms rivers and lakes. It then evaporates and forms clouds, and the cycle, which is reminiscent of the water cycle on Earth, continues.

Titan is the densest of Saturn's moons: it is a 50:50 mix of rock and water ice with a surface temperature of –292°F (–180°C). It is a gloomy world because the smog blocks 90 percent of the incident sunlight. Cassini revealed that its surface is shaped by Earth-like processes—tectonics, erosion, and winds—and perhaps ice volcanism. No liquid methane was detected on the initial flybys, but drainage channels and dark elliptical regions, thought to be evaporated lakes, showed where fluids had been. Linear features nicknamed "cat scratches" were also identified.



TITAN'S ATMOSPHERE
Infrared and ultraviolet data combined reveals aspects of the atmosphere. Areas where methane absorbs light appear orange and green. The high atmosphere is blue.



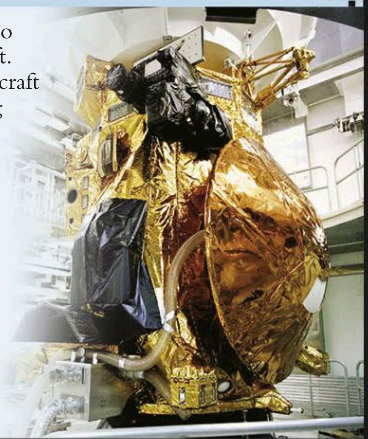
ORANGE AND PURPLE HAZE
The upper atmosphere consists of separate layers of haze; up to 12 have been detected in this ultraviolet, true-color image of Titan's night-side limb.

ICY SURFACE
The surface of Titan is shown in this image taken by the Huygens lander in 2005. The pebbles in the foreground are up to about 6 in (15 cm) across and are thought to be composed of frozen water.

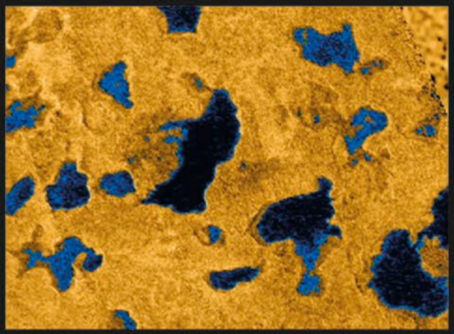
EXPLORING SPACE

THE HUYGENS PROBE

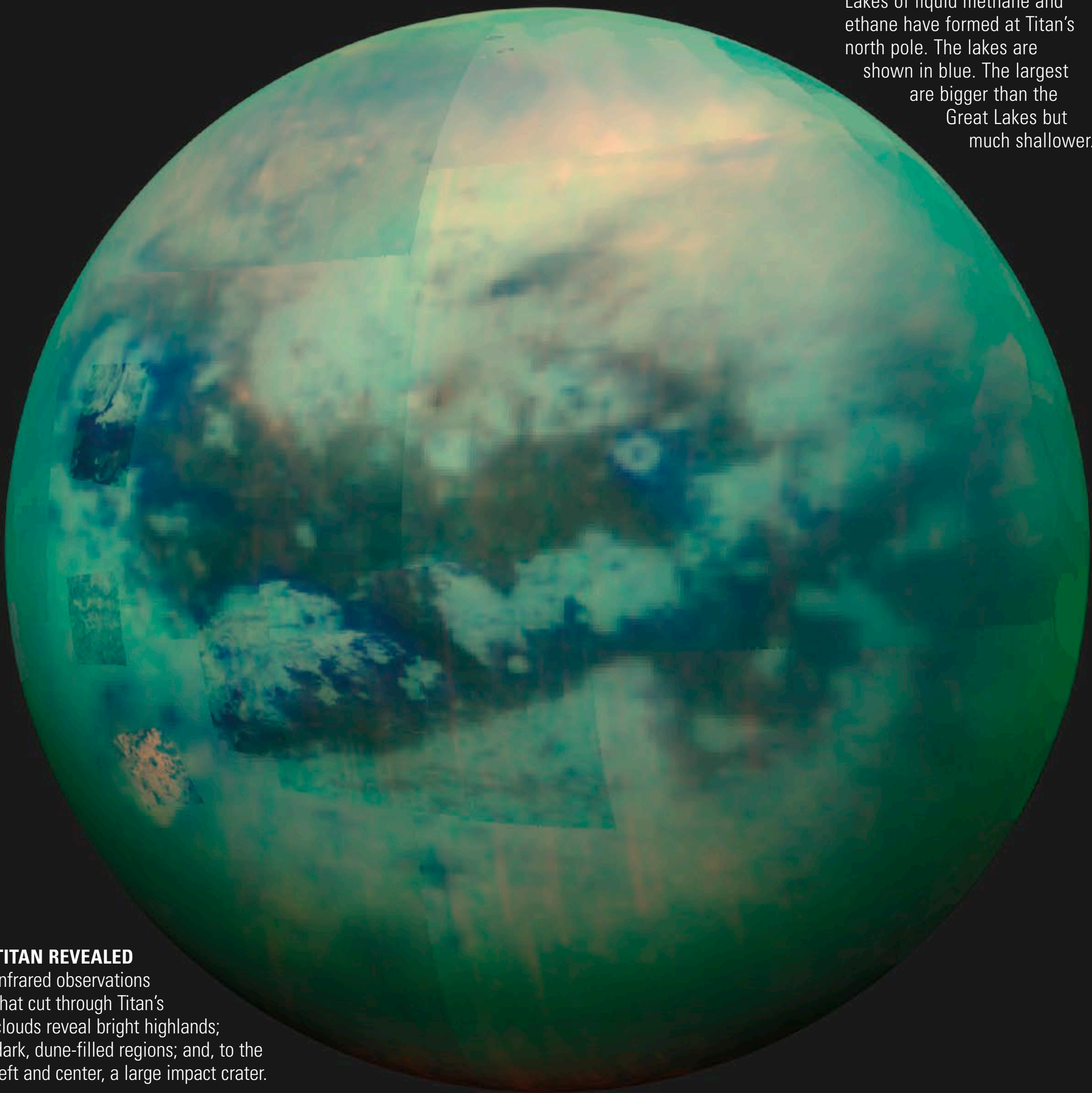
The European probe Huygens traveled to Titan onboard NASA's Cassini spacecraft. Once there, it separated from the larger craft and parachuted into Titan's haze. During its 2.5-hour descent, Huygens tested the atmosphere, measured the speed of the buffeting winds, and took images of the moon's surface. An instrument recorded the first surface touch on January 14, 2005, sending back evidence of a thin, hard crust with softer material beneath.



HUYGENS AND CASSINI
The shield-shaped Huygens probe (right) is attached to Cassini's frame in preparation for the launch from Cape Canaveral, Florida, in October 1997.



POLAR LAKES
Lakes of liquid methane and ethane have formed at Titan's north pole. The lakes are shown in blue. The largest are bigger than the Great Lakes but much shallower.



TITAN REVEALED
Infrared observations that cut through Titan's clouds reveal bright highlands; dark, dune-filled regions; and, to the left and center, a large impact crater.